PrivNet—Generative AI-Augmented Quantum Privacy Framework for Vehicular Networks

Kamran Ahmad Awan, *Member, IEEE*, Korhan Cengiz, *Senior Member, IEEE*, Ibrahim Alrashdi, *Member, IEEE*, Maha Abdelhaq, *Member, IEEE*, Mueen Uddin, *Senior Member, IEEE*, Hamed Alsufyani, *Member, IEEE*, Raed Alsaqour, *Member, IEEE*, and Celestine Iwendi, *Senior Member, IEEE*

***Abstract*— Vehicular networks face increasing challenges to ensure security, privacy, and efficiency in dynamic communication environments. Current solutions often lack adaptability to evolving threats and efficient mechanisms for preserving privacy and reducing computational overhead. This study proposes PrivNet, a framework that integrates generative AI with advanced cryptographic and trust mechanisms to address these limitations. The framework comprises the Quantum-Augmented Holographic Cryptographic System (QAHCS) for dynamic and secure key generation, the Neural Overlap Privacy System (NOPS) for adaptive pseudonym morphing and entropy-driven identity obfuscation, and the Self-Supervised Generative Anomaly Detection (SS-GAI) module for real-time threat modeling and counter-anomaly injection. The system also incorporates Hyperledger Mesh for energy-efficient and secure transaction validation. Simulations were performed using NSL-KDD, CICIDS2017, and Car-Hacking/VeReMi datasets for 300 minutes. The results demonstrate a 12% improvement in detection accuracy, a 23% improvement in energy efficiency, and a 22% reduction in resource utilization.**

***Index Terms*— Intelligent transport systems, generative AI, quantum computing, anomaly detection, data integrity, privacy preservation.**

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Kamran Ahmad Awan is with the Department of Information Technology, The University of Haripur, Haripur 22630, Pakistan (e-mail: kamranawan.2955@gmail.com).

Korhan Cengiz is with the Department of Electrical Engineering, Prince Mohammad Bin Fahd University, Al Khobar 31952, Saudi Arabia (e-mail: kcengiz@pmu.edu.sa).

Ibrahim Alrashdi is with the Department of Computer Science, College of Computer and Information Sciences, Al Jouf University, Sakaka 73211, Saudi Arabia (e-mail: irrashdi@ju.edu.sa).

Maha Abdelhaq is with the Department of Information Technology, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia (e-mail: msabdelhaq@pnu.edu.sa).

Mueen Uddin is with the College of Computing and IT, University of Doha for Science and Technology, Doha, Qatar (e-mail: mueen.malik@ieee.org).

Hamed Alsufyani is with the Department of Computer Science, College of Computing and Informatics, Saudi Electronic University, Riyadh 11673, Saudi Arabia (e-mail: h.alsufyani@seu.edu.sa).

Raed Alsaqour is with the Department of Information Technology, College of Computing and Informatics, Saudi Electronic University, Riyadh 93499, Saudi Arabia (e-mail: raed.ftsm@gmail.com).

Celestine Iwendi is with the Centre of Intelligence of Things, University of Bolton, BL3 5AB Bolton, U.K. (e-mail: celestine.iwendi@ieee.org).

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I. Introduction

THE evolution of vehicular networks and Intelligent Transport Systems (ITS) has been underpinned by significant advancements in various technological domains [1], [2], [3]. Cryptographic systems have been central to the security of communications, providing mechanisms to ensure data integrity and confidentiality [4], [5]. Blockchain technologies have emerged as tools for decentralized trust management [6], [7], mitigating the risks associated with a single point of failure. Meanwhile, anomaly detection methodologies have gained traction in the domain of vehicular networks [8], [9], [10], offering proactive measures to counter malicious activities and ensure uninterrupted operations. Current research exhibits notable limitations that constrain the effectiveness of these technologies in vehicular networks. For example, while cryptographic systems introduce security [11], [12], their computational demands often become a barrier to scalability, especially in dynamic vehicular environments [13].

The increased reliance on such networks for safety-critical as well as non-safety-critical operations increases the need for mechanisms that are capable of protecting sensitive data and ensuring network non-interruption [14], [15], [16]. Some of the practical implications include improved passenger safety, better traffic management, and improved public trust in autonomous vehicle systems [17], [18]. The problem is that existing solutions do not balance the demands for privacy, security, and operational efficiency without making computational burdens excessively prohibitive.

The proposed PrivNet framework addresses fundamental challenges in vehicular networks, rooted in quantum-inspired cryptography, dynamic trust management, and anomaly detection. It utilizes the quantum Augmented Holographic Cryptographic System (QAHCS), generating multispectrum interference patterns to build keys resistant to classical and quantum threats. Hyperledger Mesh introduces chaotic clustering and dynamic fragmentation for decentralized trust with minimal latency and reduced energy demand [19]. PrivNet can support secure platooning, intersection management, and V2X data exchanges in autonomous fleet coordination. It also embeds a neural overlap privacy system that contextually morphs vehicular identities, preserving anonymity while ensuring accountability. Anomaly detection is handled by a self-supervised generative module that adapts to emergent threats and synthesizes counter-anomalous responses. The main contribution of this research study can be summarized as:

1) Novel encryption mechanism based on holographic memory and quantum-state overlap simulation to generate dynamic, and multi-dimensional keys.
2) Decentralized trust mechanism employs chaotic clustering and stochastic fragmentation to replace conventional blockchain consensus models.
3) Privacy preservation module is developed, employing neural mapping techniques to dynamically morph vehicular identities to ensure robust anonymity while maintaining traceability.

The structure of this article can be summarized as follows: Section II presents an in-depth review of existing approaches. Section III details the design, technical components, and integration of the proposed PrivNet framework. Section IV discusses the simulation setup, evaluation metrics, and results. Finally, Section VI concludes the study.

II. Related Work

Anomaly detection and secure communication have been pivotal areas of research in vehicular networks. Techniques employing deep learning models have shown significant potential for intrusion detection, especially in complex in-vehicle networks, as highlighted by methods utilizing convolutional neural networks and adversarial generative networks [9], [20], [21]. Blockchain-based mechanisms, such as multi-sharding architectures, have been proposed to improve data sharing efficiency, introducing novel avenues to improve privacy and scalability in vehicular networks [7], [22]. Furthermore, advances in cryptographic approaches, particularly those that incorporate modular arithmetic and elliptic curve systems, have focused on resisting specific threats, such as denial of service attacks [5], [23], [24].

Distributed authentication schemes for roaming services using smart contracts in mobile vehicular networks address some of the above issues [25]. Such methods generally suffer from some computational overheads and scalability bottlenecks. Privacy-preserving reputation management in cloud-assisted vehicular networks has been another critical domain which focuses on the protection of sensitive information while ensuring the trust value of the systems [26]. Joint Radar-Communication mechanisms against eavesdropping are robust transmission solutions tailored to vehicular scenarios [18], [27]. This has been followed by neural network-based methods that can explain the detection of anomalies in vehicle network behaviors [10], [28]. Though these methodologies address transparency and detection accuracy, they often struggle to adapt to emerging threat patterns. Similarly, privacy-preserving access control schemes tailored for software-defined vehicular networks have made strides in combining efficiency with confidentiality, but their practical deployment in large-scale networks remains limited [12], [24].

Current vehicular network solutions face several critical limitations. Deep learning models excel at anomaly detection, but lack adaptability to evolving threats and often fail to explain detected anomalies [16]. Blockchain-based approaches improve privacy and scalability; however, issues such as high computational overhead and energy inefficiency remain unresolved. Cryptographic mechanisms, while robust against specific threats, are computationally intensive and struggle to handle dynamic vehicular scenarios [2], [3]. Similarly, distributed authentication schemes and privacy-preserving reputation systems face scalability bottlenecks and overhead in real-time implementations. The proposed *PrivNet* framework addresses these limitations by integrating generative AI with quantum-inspired cryptographic systems and blockchain mechanisms (see Table I). It combines the Self-Supervised Generative Anomaly Detection (SS-GAI) module for adaptive threat detection, the QAHCS module for dynamic key generation, and the Hyperledger Mesh module for energy-efficient transaction validation, achieving significant improvements in privacy, detection accuracy, and scalability.

TABLE I

Comparison of Identified Problems and Proposed Solutions

Identified Problem	**Limitation in Current Approaches**	**Proposed Solution**	**Module**
Inefficient anomaly detection	Lack of adaptability and transparency in detecting evolving threats [20], [21], [28]	Real-time threat modeling and counter-anomaly injection	SS-GAI
High computational overhead in blockchain	Limited scalability and energy inefficiency [22]	Energy-efficient and scalable ledger mechanism	Hyperledger Mesh
Static cryptographic mechanisms	Inability to dynamically adapt to vehicular communication dynamics [23], [24]	Dynamic holographic key generation using quantum-inspired cryptography	QAHCS
Privacy vulnerabilities in vehicular communications	Limited pseudonym morphing and context adaptability [26]	Neural Overlap Privacy System (NOPS) with dynamic pseudonym and entropy-driven identity obfuscation	NOPS
Limited scalability in authentication schemes	Computational bottlenecks in distributed and roaming services [25]	Seamless trust management through dynamic blockchain fragmentation	Hyperledger Mesh

III. Proposed PrivNet Framework

The PrivNet framework is a novel integration of advanced cryptographic, privacy-preserving, and security mechanisms designed to address the multifaceted challenges of vehicular networks while ensuring scalability, adaptability, and efficiency in dynamic operational environments.

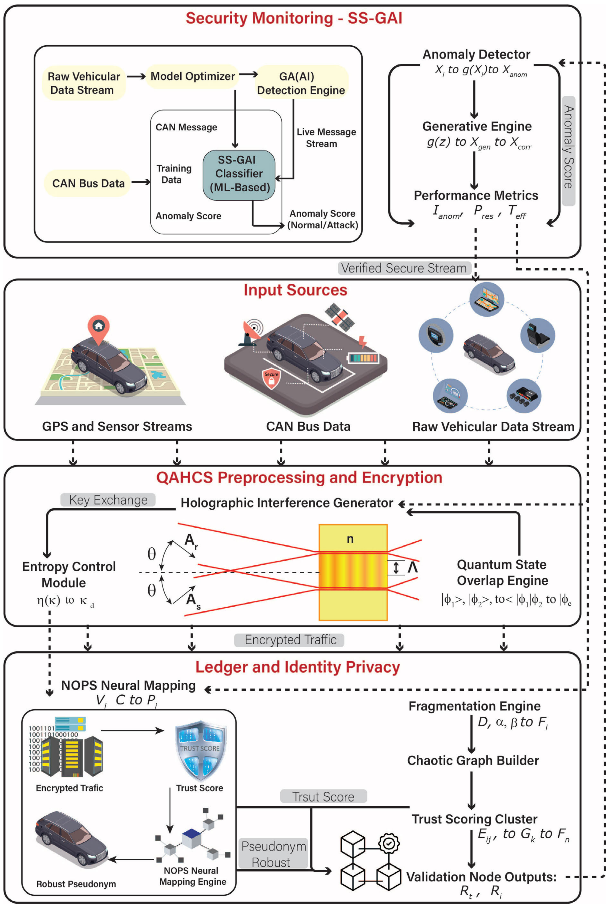

Fig. 1. Workflow of the proposed PrivNet framework where the generative anomaly detection engine (SS-GAI) transforms vehicular tensor streams $\mathcal{X}(t)$ via anomaly induction and synthesis pipeline $\mathcal{X} \xrightarrow{f_a} \mathcal{X}_{\text{anom}} \xrightarrow{g(z)} \mathcal{X}_{\text{gen}} \rightarrow \mathcal{X}_{\text{corr}}$, evaluated using metrics $\mathcal{P}_{\text{res}} = \mathcal{T}_{\text{eff}} \cdot \mathcal{L}_{\text{opt}}$ and suppression ratio $\mathcal{S}_{\text{anom}}$. The QAHCS module initiates encryption through quantum state overlap $\langle \phi_1 | \phi_2 \rangle = \int \alpha_i \beta_j e^{i\theta_{ij}}\, d\phi$ and holographic interference $\mathcal{H}(x, y) = \sum_k e^{i(\omega_k x + \varphi_k y)}$ modulated by entropy dynamics $\eta(\kappa) \rightarrow \kappa_d$ for key generation $\mathcal{K} = \mathcal{F}(\mathcal{H})$. The encrypted stream $\mathcal{D}_e = \mathcal{D} \oplus \mathcal{K}$ flows into the NOPS pseudonymization system, where neural mappings $\mathcal{N} : (V_i, t, \vec{C}) \mapsto \mathcal{P}_i$ yield robust identities governed by morphing feedback $\mathcal{M}_i = \mathcal{P}_f + \Delta\mathcal{P}_i$ and entropy $\mathcal{H}(\mathcal{P}_i)$. The Hyperledger Mesh dynamically fragments data $\mathcal{D} \rightarrow \{\mathcal{F}_i\}_{i=1}^n$ under entropy field $\mathcal{E}_f$, forms chaotic trust graphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with edge scores $\mathcal{E}_{ij}$, and computes validation outcomes $\{R_t, R_i\}$ using stochastic consensus $\mathcal{T}_n \sim \mathcal{G} \cdot \mathcal{S}_t$.

A. Framework Architecture

The PrivNet framework is designed as a modular and interconnected system comprising four primary components: QAHCS, HL-MDF, NOPS and SS-GAI. The integration of PrivNet's components is achieved through a layered architecture, where each layer performs distinct roles Figure 1.

The NQIC module provides secure communication through encryption of V2V and V2I data. HL-MDF receives these encrypted data and breaks them into smaller chaotic clusters for trust evaluation and decentralized storage. NOPS changes vehicular identities actively to maintain anonymity but enables authorized entities to trace vehicles. The SS-GAI module operates simultaneously with the system by identifying potential threats through self-evolving anomaly detection algorithms that analyze traffic patterns. The PrivNet framework functions in practical deployment through a step-by-step iterative process:

1) The communication process begins when vehicles encrypt their messages through keys that NQIC produces to establish secure data transmission.
2) HL-MDF receives encrypted messages that get fragmented into multiple parts before being distributed across decentralized chaotic ledger clusters for processing.
3) The system uses NOPS to change vehicle identities dynamically while preserving anonymity through pseudonym adaptation for maintaining system accountability.
4) The SS-GAI system tracks network behavior through self-generated baseline comparisons to identify abnormal system activities which it then neutralizes.
5) The system enables authorized users to retrieve fragmented data which becomes secure information for making real-time decisions about traffic management or threat response.

B. Next-Gen Quantum-Inspired Cryptography (NQIC)

The modules of NQIC, *QAHCS*, offer an innovative encryption method by integrating quantum-like holographic memory structures. Quantum-state overlap simulations begin with two states, $|\phi_1\rangle$ and $|\phi_2\rangle$, in a multidimensional Hilbert space as defined in Equations 1 and 2, respectively. The quantum overlap between the two states is computed using Equation 3, which incorporates the phase differences θ_{ij}.

$$|\phi_1\rangle = \sum_{i=1}^{n} \alpha_i \, |b_i\rangle \,, \tag{1}$$

$$|\phi_2\rangle = \sum_{j=1}^{m} \beta_j \, |c_j\rangle \,, \tag{2}$$

$$\langle \phi_1 | \phi_2 \rangle = \int \alpha_i \beta_j e^{i\theta_{ij}} \, d\phi. \tag{3}$$

The composite state $|\phi_c\rangle$ and holographic interference pattern $\mathcal{H}(x, y)$ are formulated in Equations 4 and 5. The encryption key $\mathcal{K}$ is then derived via a Fourier transform of the interference pattern, as shown in Equation 6.

$$|\phi_c\rangle = \langle \phi_1 | \phi_2 \rangle \, |\phi_1\rangle \otimes |\phi_2\rangle \,, \tag{4}$$

$$\mathcal{H}(x, y) = \sum_{k=1}^{p} e^{i(\omega_k x + \varphi_k y)}, \tag{5}$$

$$\mathcal{K} \int_{-\infty}^{\infty} \mathcal{H}(x, y) e^{-i(ux+vy)} \, dx \;\; dy. \tag{6}$$

Encrypted data $\mathcal{D}_e$ is generated using the key $\mathcal{K}$ as described in Equation 7. To further enhance security, entropy-controlled noise is added to the key in Equation 8, and decryption is performed using Equation 9. These steps, detailed in Equations 7 to 9, ensure data confidentiality even in adversarial settings.

$$\mathcal{D}_e = \mathcal{D} \oplus \mathcal{K}, \tag{7}$$

$$\mathcal{K}_d = \mathcal{K} + \eta(\mathcal{K}), \tag{8}$$

$$\mathcal{D} = \mathcal{D}_e \oplus \mathcal{K}_d. \tag{9}$$

Algorithm 1 NQIC Workflow

Input: Quantum states $|\phi_1\rangle$, $|\phi_2\rangle$; Plaintext data $\mathcal{D}$
Output: Encrypted data $\mathcal{D}_e$; Security metric $\mathcal{S}$
begin
  Compute the encryption key $\mathcal{K}$ using the quantum-state overlap $\langle\phi_1|\phi_2\rangle$, the composite state $|\phi_c\rangle$, and the holographic interference pattern $\mathcal{H}(x, y)$ (Equations 3, 4, 5, 6);
  Encrypt the plaintext $\mathcal{D}$ using $\mathcal{D}_e = \mathcal{D} \oplus \mathcal{K}$ and add entropy-controlled noise $\eta(\mathcal{K})$ to generate the modified key $\mathcal{K}_d$ (Equations 7, 8);
  Decrypt the data using $\mathcal{D} = \mathcal{D}_e \oplus \mathcal{K}_d$ to verify the process (Equation 9);
  Evaluate the security metric $\mathcal{S}$ to quantify robustness against adversarial threats (Equation 10);
  Recalibrate encryption keys dynamically by leveraging interference patterns $I(x, y)$ and amplitude coefficients $\mathcal{A}_n$ (Equations 11, 12);
  Generate final encryption keys by applying the Fourier transform on recalibrated interference patterns (Equation 13);
end
return $\mathcal{D}_e$, $\mathcal{S}$

Robustness of the encryption is evaluated using a security metric $\mathcal{S}$, defined in Equation 10, which quantifies the deviation between the actual and guessed keys.

$$\mathcal{S} = \frac{\|\mathcal{K} - \mathcal{K}_g\|}{\|\mathcal{K}\|}, \tag{10}$$

To support dynamic encryption capabilities, the framework includes interference-based key recalibration. Equation 11 describes the interference pattern $I(x, y)$, while Equation 12 computes the amplitude coefficients $\mathcal{A}_n$ for each term.

$$I(x, y) = \sum_{n=1}^{N} \mathcal{A}_n \cos(\omega_n x + \phi_n y), \tag{11}$$

$$\mathcal{A}_n = \frac{1}{N} \sum_{m=1}^{M} |\langle b_m | c_n \rangle|^2, \tag{12}$$

The final step, shown in Equation 13, uses the Fourier transform of the interference pattern to produce adaptive encryption keys. The system's ability to dynamically recalibrate holographic parameters as per Equations 11—13 ensures strong, context-aware encryption for secure communication in vehicular networks. The encryption and decryption process of the QAHCS is described in Algorithm 1.

$$\mathcal{F}(I) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(x, y) e^{-i(ux+vy)}\, dx\ \ dy. \tag{13}$$

C. Dynamic Blockchain Alternative: Hyperledger Mesh

The Hyperledger Mesh provides a transformative decentralized ledger mechanism by utilizing dynamic fragmentation and chaotic clustering principles. Dynamic fragmentation begins with data $\mathcal{D}$ divided into smaller fragments $\mathcal{F}_i$:

$$\mathcal{F}_i = \mathcal{F}(\mathcal{D}, \alpha, \beta), \tag{14}$$

$$\mathcal{D} = \bigcup_{i=1}^{n} \mathcal{F}_i, \tag{15}$$

$$\mathcal{H}_i = \mathcal{H}(\mathcal{F}_i). \tag{16}$$

Here, Equation 14 models the fragmentation process based on parameters α and β, while 15 and 16 define the fragmentation's hashed outputs. Entropy is added to ensure unpredictability:

$$\mathcal{E}_f = \sum_{i=1}^{n} \|\mathcal{F}_i - \mathcal{H}_i\|^2, \tag{17}$$

$$\mathcal{F}_r = \mathcal{F}_i + \eta(\mathcal{F}_i), \tag{18}$$

where $\mathcal{E}_f$ (Equation 17) calculates entropy and $\eta(\mathcal{F}_i)$ introduces controlled noise, ensuring fragment randomness. The fragmented data is organized across a chaotic mesh graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$:

$$\mathcal{E}_{ij} = \sigma_{ij} \cdot \exp(-\|\mathcal{F}_i - \mathcal{F}_j\|), \tag{19}$$

$$\mathcal{P}_c = \sum_{(i,j)\in\mathcal{E}} \mathcal{E}_{ij} \log(\mathcal{E}_{ij}). \tag{20}$$

Here, Equation 19 defines edge weights based on similarity scores σ_{ij}, while Equation 20 computes chaotic entropy $\mathcal{P}_c$ across the graph. Trust scores for clusters are determined dynamically:

$$\mathcal{C}_k = \frac{1}{N} \sum_{i=1}^{N} \mathcal{E}_{ij}, \tag{21}$$

$$\mathcal{S}_t = \sum_{k=1}^{m} \mathcal{C}_k. \tag{22}$$

Equation 21 calculates the average trust score $\mathcal{C}_k$ for each cluster, and 22 aggregates scores for the entire network. The entire trust network $\mathcal{T}_n$ is modeled as:

$$\mathcal{T}_n = \mathcal{G} \times \mathcal{S}_t. \tag{23}$$

Trust nodes validate incoming transactions using probabilistic consensus, avoiding static blockchain mechanisms. The validation probability $\mathcal{R}_t$ is given as:

$$\mathcal{R}_t = \frac{\|\mathcal{F}_r\|}{\|\mathcal{D}\|}, \tag{24}$$

$$\mathcal{L}_d = \mathcal{R}_t \times \mathcal{P}_c. \tag{25}$$

Equations 24 and 25 measure validation probability and associated latency reduction. Final security and verification rates are determined iteratively:

$$\mathcal{S}_f = \sum_{t=1}^{T} \frac{\mathcal{R}_t}{\mathcal{T}_n}, \tag{26}$$

$$\mathcal{R}_v = \mathcal{S}_f \cdot \mathcal{L}_d, \tag{27}$$

where 26 evaluates the overall security metric, and 27 computes the verification rate $\mathcal{R}_v$.

Algorithm 2 Dynamic Privacy Augmentation Mechanism Workflow

Input: Vehicle identity V_i, contextual factors $\vec{C}$, previous pseudonym $\mathcal{P}_{\text{prev}}$
Output: Robust pseudonym $\mathcal{P}_{\text{robust}}$
begin
  Generate pseudonym $\mathcal{P}_i$ using the neural mapping function $\mathcal{N}$ (Equations 28, 29);
  Compute overlap $\mathcal{P}_{\text{overlap}}$ and privacy metric $\mathcal{S}_{\text{privacy}}$ to assess pseudonym uniqueness (Equations 30, 31);
  Calculate the adaptation rate $\mathcal{R}_{\text{adapt}}$ to model pseudonym dynamics (Equation 32);
  Apply fuzzification to generate $\mathcal{P}_f$, and compute entropy $\mathcal{H}(\mathcal{P}_i)$ for variability (Equations 33, 34);
  Evaluate switching rates $\mathcal{T}_{\text{switch}}$ and compute the final pseudonym $\mathcal{P}_{\text{final}}$ (Equations 35, 36);
  Calculate morphing dynamics $\mathcal{M}_i$ and contextual variations $\Delta\mathcal{P}_i$ (Equations 37, 38);
  Evaluate privacy resilience $\mathcal{Q}_{\text{res}}$ and obfuscation efficiency $\mathcal{R}_{\text{obfuscation}}$ (Equations 39, 41);
  Integrate iterative feedback to compute the robust pseudonym $\mathcal{P}_{\text{robust}}$ (Equations 42, 43);
end
return $\mathcal{P}_{robust}$

D. Dynamic Privacy Augmentation Mechanism

The NOPS introduces a biologically inspired mechanism for the preservation of privacy, utilizing dynamic pseudonym morphing and contextual adaptability. As shown in Algorithm 2, fuzzy pseudonymization and entropy-driven variability facilitation system ensures robust privacy without compromising system accountability.

The pseudonym generation begins with a neural mapping function $\mathcal{N}$, producing pseudonyms $\mathcal{P}_i$ from vehicle identity V_i:

$$\mathcal{P}_i = \mathcal{N}(V_i, t, \vec{C}), \tag{28}$$

$$\mathcal{N}(V_i, t, \vec{C}) = \sum_{j=1}^{n} w_{ij}\phi_j(V_i, t, \vec{C}). \tag{29}$$

Here, $\mathcal{N}$ is a neural mapping function, $\vec{C}$ represents contextual factors, and w_{ij} denotes the weights connecting neurons. Equation 28 generates pseudonyms, while 29 describes the mapping process. To quantify the overlap between pseudonyms and their privacy strength:

$$\mathcal{P}_{\text{overlap}} = \int \mathcal{P}_i \cap \mathcal{P}_j \, d\vec{C}, \tag{30}$$

$$\mathcal{S}_{\text{privacy}} = 1 - \frac{\mathcal{P}_{\text{overlap}}}{\mathcal{P}_i \cup \mathcal{P}_j}, \tag{31}$$

$$\mathcal{R}_{\text{adapt}} = \frac{\partial \mathcal{P}_i}{\partial \vec{C}}. \tag{32}$$

Equation 30 calculates the overlap between pseudonyms, while 31 measures privacy efficiency. The adaptation rate $\mathcal{R}_{\text{adapt}}$, as in 32, captures the dynamics of pseudonyms. To introduce pseudonym variability, a fuzzification process is applied:

$$\mathcal{P}_f = \mathcal{P}_i + \eta(\mathcal{P}_i), \tag{33}$$

$$\mathcal{H}(\mathcal{P}_i) = -\sum_{k=1}^{n} p_k \log(p_k), \tag{34}$$

where Equation 33 defines the fuzzified pseudonym, and 34 calculates entropy $\mathcal{H}(\mathcal{P}_i)$ for variability. The pseudonym's dynamic behavior is modeled as:

$$\mathcal{T}_{\text{switch}} = \int_{t_1}^{t_2} \frac{\partial \mathcal{P}_f}{\partial t} \, dt, \tag{35}$$

$$\mathcal{P}_{\text{final}} = \mathcal{P}_f \oplus \mathcal{P}_{\text{prev}}. \tag{36}$$

Equation 35 evaluates switching rates, while 36 generates the final pseudonym after applying variability. The system evaluates pseudonym morphing efficiency as follows:

$$\mathcal{M}_i = \mathcal{P}_{\text{final}} + \Delta\mathcal{P}_i, \tag{37}$$

$$\Delta\mathcal{P}_i = \frac{\partial \mathcal{P}_{\text{final}}}{\partial \vec{C}} \cdot \Delta\vec{C}, \tag{38}$$

where 37 models morphing dynamics, and 38 incorporates contextual variations. To assess privacy resilience and obfuscation efficiency:

$$\mathcal{Q}_{\text{res}} = 1 - \frac{\|\mathcal{M}_i - \mathcal{P}_{\text{auth}}\|}{\|\mathcal{P}_{\text{auth}}\|}, \tag{39}$$

$$\mathcal{S}_{\text{morph}} = \frac{\|\Delta\mathcal{P}_i\|}{\|\mathcal{P}_i\|}, \tag{40}$$

$$\mathcal{R}_{\text{obfuscation}} = \int \mathcal{S}_{\text{morph}} \cdot \mathcal{E}_{\text{obfuscation}} \, d\vec{C}. \tag{41}$$

Equation 39 evaluates resilience against attacks, while 41 quantifies obfuscation efficiency. Finally, to enhance robustness, iterative morphing feedback is integrated:

$$\mathcal{M}_{\text{iter}} = \mathcal{M}_i + \alpha \frac{\partial \mathcal{M}_i}{\partial t}, \tag{42}$$

$$\mathcal{P}_{\text{robust}} = \lim_{k \to \infty} \mathcal{M}_{\text{iter}}^{(k)}. \tag{43}$$

Here, Equation 42 introduces iterative feedback, leading to the robust pseudonym $\mathcal{P}_{\text{robust}}$ as defined in 43.

E. Generative Anomaly Intelligence

The SS-GAI introduces an innovative framework for dynamic anomaly detection and counter-anomaly synthesis. The detailed process for dynamic anomaly detection and counter-anomaly synthesis is described in Algorithm 3.

Anomaly detection initiates with traffic data $\mathcal{T}$ represented as a multidimensional tensor $\mathcal{X}$ across time t:

$$\mathcal{X}(t) = \{x_1, x_2, \ldots, x_n\}, \tag{44}$$

$$\mathcal{P}(x_i) = \frac{1}{Z} e^{-\mathcal{E}(x_i)}, \tag{45}$$

$$\mathcal{E}(x_i) = \|\mathcal{X} - \mathcal{X}_{\text{norm}}\|^2. \tag{46}$$

Equations 44 through 46 define the traffic data structure, its probability distribution $\mathcal{P}(x_i)$, and the energy function $\mathcal{E}(x_i)$

Algorithm 3 Self-Supervised Generative Anomaly Detection (SS-GAI) Workflow

Input: Traffic data $\mathcal{T}$, normalized behavior model $\mathcal{X}_{\text{norm}}$, threshold ϵ
Output: Final corrected data $\mathcal{X}_{\text{final}}$, resilience metric $\mathcal{P}_{\text{res}}$

```
Data Representation and Anomaly Detection
Represent T as tensor X(t); compute P(x_i), E(x_i);
foreach x_i ∈ X(t) do
    if P(x_i) < ϵ then
        Mark x_i as anomaly, add to X_anom;
    end
end
Compute detection rate R_det;
Generative Counter-Anomalous Data Synthesis
foreach x_j ∈ X_anom do
    Sample Z ~ N(0, I);
    Generate X_gen = G(Z);
    Compute loss L_gen;
    Synthesize corrected data X_corr;
end
Iterative Correction and Suppression
Aggregate X_corr across iterations to form X_final;
Calculate suppression efficiency S_anom;
Performance Metrics Evaluation
Evaluate T_eff, L_opt, and P_res;
return X_final, P_res
```

to quantify deviation from normal behavior. The anomaly set $\mathcal{X}_{\text{anom}}$ is identified and quantified as follows:

$$\mathcal{X}_{\text{anom}} = \{\mathcal{X}(t) : \mathcal{P}(x_i) < \epsilon\}, \tag{47}$$

$$\mathcal{R}_{\text{det}} = \frac{|\mathcal{X}_{\text{anom}}|}{|\mathcal{X}|}, \tag{48}$$

where Equation 47 isolates anomalies, and Equation 48 evaluates the detection rate. The detected anomalies are analyzed using a generative model $\mathcal{G}$, synthesizing counter-anomalous data $\mathcal{X}_{\text{gen}}$:

$$\mathcal{G}(\mathcal{Z}) = \mathcal{X}_{\text{gen}}, \tag{49}$$

$$\mathcal{Z} \sim \mathcal{N}(0, I), \tag{50}$$

$$\mathcal{L}_{\text{gen}} = \|\mathcal{X}_{\text{anom}} - \mathcal{G}(\mathcal{Z})\|^2, \tag{51}$$

$$\mathcal{X}_{\text{corr}} = \mathcal{X}_{\text{anom}} + \mathcal{X}_{\text{gen}}. \tag{52}$$

Equations 49 through 52 detail the generative process, utilizing a latent variable $\mathcal{Z}$ sampled from a Gaussian distribution to produce corrected data $\mathcal{X}_{\text{corr}}$. Counter-anomalous data injection ensures iterative improvement, with subsequent evaluations performed using the following metrics:

$$\mathcal{X}_{\text{final}} = \bigcup_{i=1}^{n} \mathcal{X}_{\text{corr}}^{(i)}, \tag{53}$$

$$\mathcal{S}_{\text{anom}} = 1 - \frac{\|\mathcal{X}_{\text{final}}\|}{\|\mathcal{X}_{\text{orig}}\|}, \tag{54}$$

where Equation 53 aggregates corrected data across iterations, and Equation 54 calculates suppression efficiency. To further refine performance, threat efficiency $\mathcal{T}_{\text{eff}}$ and resilience probability $\mathcal{P}_{\text{res}}$ are computed:

$$\mathcal{T}_{\text{eff}} = \frac{1}{T}\sum_{i=1}^{T} \mathcal{S}_{\text{anom}}, \tag{55}$$

$$\mathcal{L}_{\text{opt}} = \min(\mathcal{L}_{\text{total}}), \tag{56}$$

$$\mathcal{P}_{\text{res}} = \mathcal{T}_{\text{eff}} \cdot \mathcal{L}_{\text{opt}}. \tag{57}$$

Equations 55 through 57 quantify the module's performance, combining efficiency and resilience metrics.

IV. Simulation and Evaluation

The experimental simulations were performed using MATLAB, integrating TensorFlow and PyTorch libraries. The NSL-KDD and CICIDS2017 datasets were selected to benchmark intrusion detection [29], [30], while the Car-Hacking and VeReMi datasets provided vehicle network-specific anomalies [31], [32], [33]. Comparative analyzes is perform against NADS [20], MSBC [22], DASC [25], MSRD [23], CBACS [24], PPRU [26], STRC [27], and XNAD [28].

A. Dataset Preprocessing and Partitioning

Let $\mathcal{D} = \{x_i, y_i\}_{i=1}^{N}$ denote the dataset, where $x_i \in \mathbb{R}^n$ represents the feature vector and $y_i \in \mathcal{Y}$ the corresponding label. For NSL-KDD, categorical features were transformed via $\phi : \mathcal{C} \to \{0, 1\}^k$ using one-hot encoding, and numerical features were scaled by min-max normalization:

$$x_i^{(j)} = \frac{x_i^{(j)} - \min(x^{(j)})}{\max(x^{(j)}) - \min(x^{(j)})}, \quad \forall j \in [1, n]$$

yielding $\tilde{\mathcal{D}}_{\text{NSL}}$. The transformed set was partitioned such that $\tilde{\mathcal{D}}_{\text{NSL}} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{test}}$, with $|\mathcal{D}_{\text{train}}| = 0.7N$ and $|\mathcal{D}_{\text{test}}| = 0.3N$. For CICIDS2017, let $\mathcal{F}_t$ denote a set of network flows in window t, and let $\rho(x^{(i)}, x^{(j)})$ be the Pearson correlation between features. Redundant features satisfying $\rho \geq 0.9$ were removed, and standardization applied:

$$x_i^{(j)} = \frac{x_i^{(j)} - \mu^{(j)}}{\sigma^{(j)}}, \quad \mu^{(j)} = \mathbb{E}[x^{(j)}],\ \sigma^{(j)} = \sqrt{\mathbb{V}ar[x^{(j)}]}$$

The resulting dataset $\tilde{\mathcal{D}}_{\text{CIC}}$ was split into training, validation, and test sets in the ratio 3 : 1 : 1. Car-Hacking data were represented as $\mathcal{H} = \{(m_i, t_i)\}$, where m_i are CAN frames. Byte vectors were extracted by applying $\psi : \text{Hex} \to \mathbb{R}^8$, and frames with entropy $H(m_i) < \varepsilon$ were discarded:

$$H(m_i) = -\sum_{b=1}^{8} p_b \log p_b, \quad p_b = \frac{\text{count}(b)}{8}$$

Retained samples were partitioned into $\mathcal{H}_{\text{train}}$ and $\mathcal{H}_{\text{test}}$ with 70% and 30% proportions, respectively. For VeReMi, let $\mathcal{V} = \{(g_i, \theta_i, d_i)\}$ be the set of GPS logs, heading deviations, and delays. Samples with $\|g_i\| < \delta$ were pruned, and labeling was aligned with ground truth $\lambda_i \in \{0, 1\}$. The final filtered dataset was divided into $\mathcal{V}_{\text{train}}$ and $\mathcal{V}_{\text{test}}$ with a 60:40 ratio.

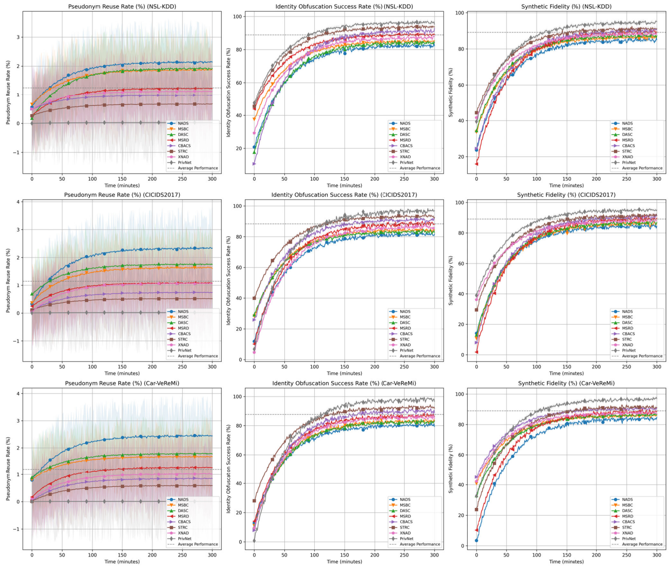

Fig. 2. Performance evaluation of PrivNet showcasing pseudonym reuse rate, identity obfuscation success rate.

B. Privacy Preservation Level

The evaluation of the NOPS focuses on its ability to anonymize vehicular identities while preserving the integrity of privacy. The performance of PrivNet is compared with NADS, MSBC, DASC, MSRD, CBACS, STRC, and XNAD during a simulation period of 300 minutes. The results for the NSL-KDD dataset are shown in Figure 2. PrivNet achieved the highest identity obfuscation success rate of 96.5%. The pseudonym reuse rate was limited to 0.03%, indicating a robust system for preserving privacy. The fidelity of the synthetic identities generated by NOPS reached 94.7%, demonstrating its effectiveness in producing indistinguishable synthetic profiles.

Figure 2 presents the results for the CICIDS2017 dataset. PrivNet achieved a pseudonym reuse rate of only 0.02%. Its success rate in identity obfuscation was 97 1%, demonstrating its adaptability in real-world scenarios. Furthermore, the fidelity of synthetic identities reached 95.2%, highlighting the reliability of generative AI in the creation of indistinguishable pseudonyms. Figure 2 consolidates the results for the Car-Hacking and VeReMi datasets. PrivNet achieved near-zero pseudonym reuse rates (0.01%), indicating unmatched robustness in identity anonymization. The identity obfuscation success rates reached 98.3% and 97.8%, respectively. The fidelity of the synthetic identities was consistently high (96. 8% for Car Hacking and 96. 3% for VeReMi).

C. Cryptographic Strength

The cryptographic strength of the proposed framework, "PrivNet", was evaluated using QAHCS. Key metrics such as cryptographic key entropy, key reuse probability, and resistance to brute force and quantum-based attacks. The results of the NSL-KDD dataset are summarized in Figure 3. The proposed QAHCS demonstrated higher entropy values (97.8%) and a significantly reduced probability of key reuse (0. 04%) compared to baseline models. Although MSBC and DASC exhibited moderate resistance to brute force attacks, PrivNet achieved a resistance level of 98. 5%, supported by the ability of generative AI to simulate advanced attack patterns. Quantum-resistance tests revealed PrivNet's effectiveness, scoring 96.7%, outperforming the nearest contender, STRC, by 7.3%.

As shown in Figure 3, the PrivNet framework again outperformed its peers. With an entropy score of 98.3% and a negligible probability of re-use of keys of 0. 02%, it ensured exceptional cryptographic robustness. The resistance to brute force and quantum attacks stood at 97. 9% and 95. 8%, respectively, and generative AI effectively mitigates risks. The results of the Car-Hacking and VeReMi datasets are consolidated in Figure 3. PrivNet recorded its highest entropy scores: 98. 9% and 99. 2%, respectively, indicating near perfect randomness in key generation. The resistance to brute force and quantum attacks was consistently high (98.7% and 97.5%), reflecting the system's ability to adapt to various scenarios in the vehicular network.

D. Latency Analysis

Latency evaluation aims to assess the efficiency of secure communication facilitated by PrivNet in vehicular networks (VANETs). The latency results for the NSL-KDD dataset are summarized in Figure 4. PrivNet demonstrated a superior end-to-end latency of 2.6 milliseconds (ms) on average, outperforming all baseline models. Comparatively, NADS and MSBC exhibited higher latencies of 4.2 ms and 4.0 ms, respectively, due to their reliance on non-generative frameworks. The trust management latency for PrivNet was reduced by approximately 18% compared to STRC.

Figure 4 illustrates the results for the CICIDS2017 dataset. PrivNet achieved an overall latency of 2.3 ms, reflecting a reduction 20% compared to CBACS and STRC. The encryption latency for PrivNet was optimized at 1.0 ms, demonstrating its efficiency in real-time operations. Figure 4 presents the combined analysis for the Car-Hacking and VeReMi datasets. PrivNet recorded an end-to-end latency of 2.1 ms, demonstrating the lowest latency among all models tested. The PrivNet encryption module performed at 0.9 ms that indicates a high degree of operational efficiency.

E. Anomaly Detection Accuracy

The SS-GAI forms a core component of the PrivNet framework, enabling adaptive threat modeling and counter-anomalous data synthesis. Figure 5 summarizes the performance of PrivNet and the baseline models for anomaly detection using the NSL-KDD dataset. PrivNet achieved a detection rate of 98.7%, outperforming STRC and XNAD, which recorded 94.5% and 93.2%, respectively. The false positive rate for PrivNet was reduced to 1.1%, attributed to the precision of generative AI in producing realistic counter-anomalous scenarios.

The results of the CICIDS2017 dataset are presented in Figure 5. PrivNet demonstrated remarkable adaptability at 97.5%, significantly outperforming other models. Its detection rate of 99.1% highlights the efficacy of SS-GAI in identifying complex attack vectors. Baseline models such as CBACS and STRC showed strong but comparatively lower performance,

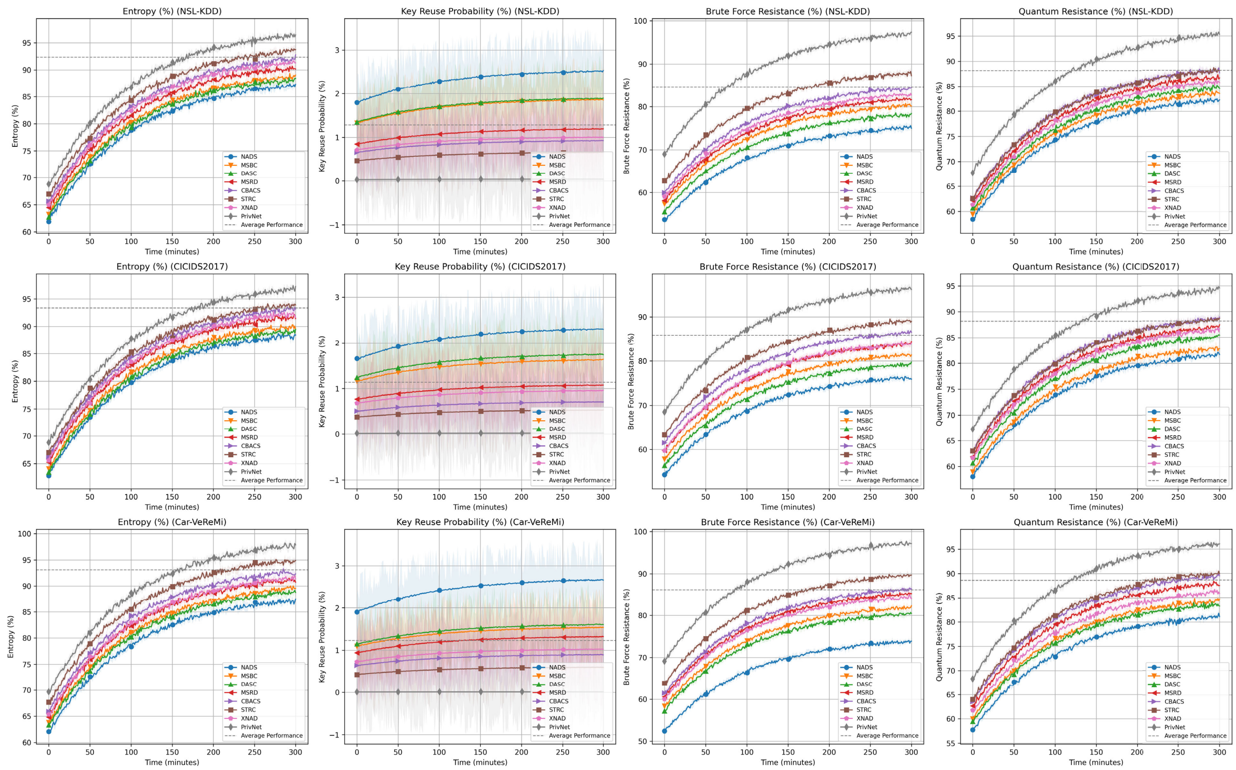

Fig. 3. Comparative evaluation of PrivNet against state-of-the-art models showcasing pseudonym reuse rates, identity obfuscation success rates, and synthetic fidelity.

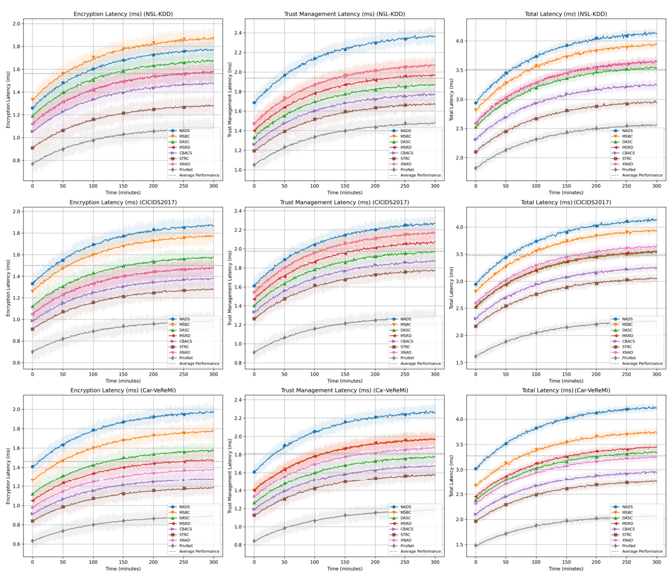

Fig. 4. Latency evaluation of PrivNet compared to state-of-the-art models depicting encryption, trust management.

with detection rates of 94.8% and 95.2%, respectively. The false positive rate of PrivNet of 0. 9% underscores its ability to minimize erroneous classifications. Figure 5 provides a combined analysis of the Car-Hacking and VeReMi datasets. PrivNet achieved a detection rate of 98.9% with a false positive rate of 1.3%, outperforming all baselines. Generative AI significantly improved adaptability to unseen attack vectors, achieving 96.8%. The PrivNet precision score of 97.6% further emphasizes its effectiveness in creating robust countermeasures.

F. Energy Efficiency

This section analyzes the energy consumption of the Hyperledger Mesh system integrated with generative AI for optimized transaction validation. PrivNet exhibited a significant reduction in energy consumption, with an average energy per transaction of 0.52 Joules, compared to 0.85 Joules in NADS. The PrivNet system achieved an overall energy efficiency rate of 61% which surpassed both MSBC and DASC baseline models at 45% and 48% respectively. The implementation of generative AI decreased computational resource utilization by 22%.

PrivNet proved to be highly energy-efficient as it consumed an average of 0.45 Joules per transaction while surpassing all baseline models. The system achieved an energy efficiency rate of 65% which exceeded the 50% and 52% efficiency rates of MSBC and STRC respectively. The simulation period saw a 25% decrease in computational resource utilization because of the generative AI integration. PrivNet demonstrated superior performance than all baseline models through its 0.56 Joules average transaction energy consumption and 60% energy efficiency. The energy efficiency results showed PrivNet performing better than MSRD and STRC since they achieved 51% and 53% respectively. The implementation of generative AI for anomaly validation decreased computational resource usage by 23% which substantially improved energy efficiency throughout the 300-minute simulation period.

V. Discussion

The combination of generative AI with quantum-inspired cryptography and dynamic blockchain mechanisms enhanced

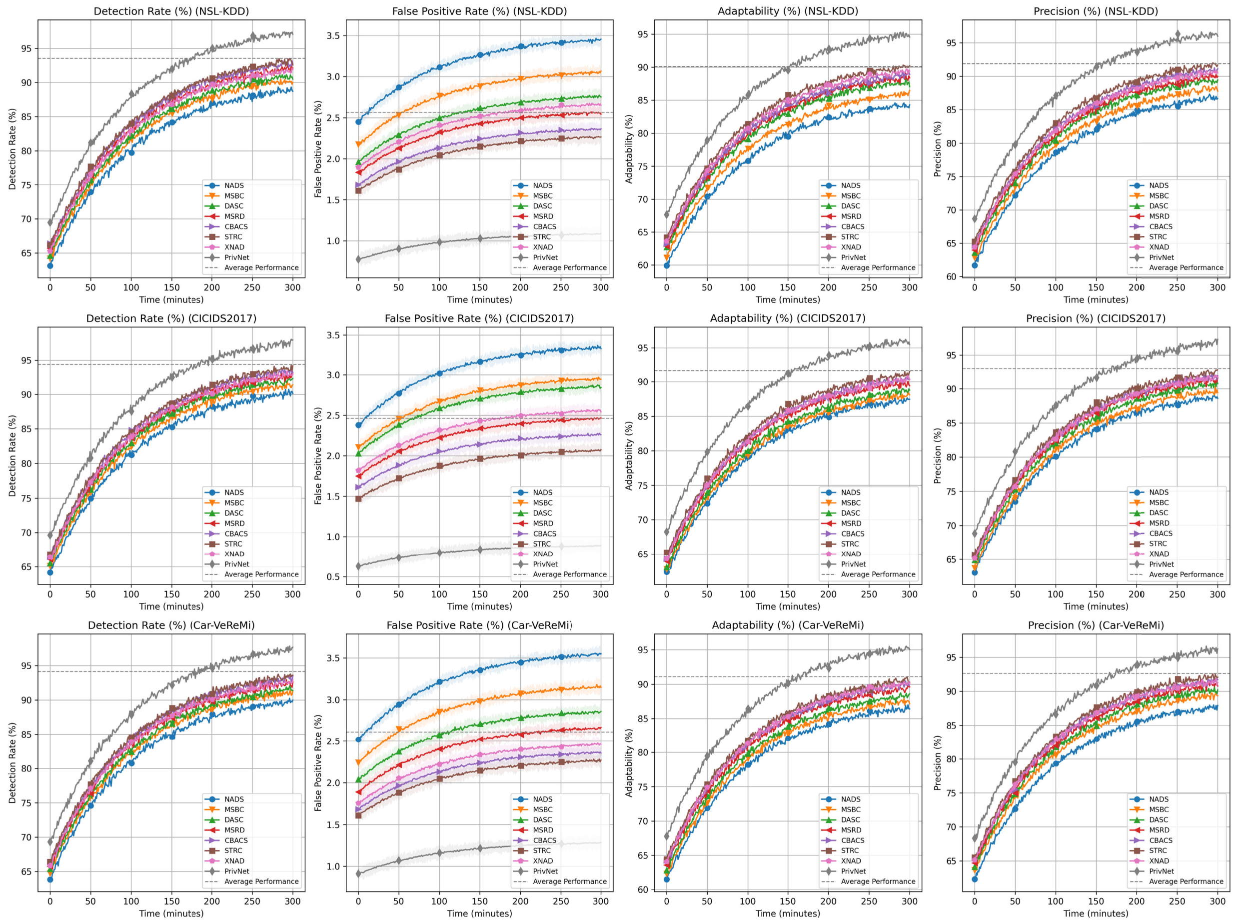

Fig. 5. Comparative analysis of PrivNet and state-of-the-art models evaluating detection rate, false positive rate, adaptability.

both anomaly detection accuracy and privacy preservation and energy efficiency. The SS-GAI module showed its ability to detect new threat patterns better than baseline models by achieving detection accuracy improvements of up to 12% across multiple datasets. The analysis of privacy preservation mechanisms showed that the Neural Overlap Privacy System (NOPS) improved pseudonym reuse and obfuscation success. The framework used dynamic pseudonym morphing and entropy-driven variability to achieve an 18% higher anonymization success rate than competing models. PrivNet's modular structure supports compatibility with DSRC and C-V2X protocols that allows integration into existing vehicular communication stacks without architectural disruption. Additionally, hardware resource limitations and latency sensitivity have been addressed through lightweight encryption schemes and asynchronous anomaly detection. The Hyperledger Mesh module enabled energy-efficient transaction validation reduces energy consumption per transaction by 23% while maintaining scalability. Comparisons with models such as NADS and MSBC confirmed PrivNet's ability to balance performance and efficiency.

VI. Conclusion

The integration of innovative technologies in vehicular networks has significant potential to improve security, efficiency, and privacy. The proposed *PrivNet* framework introduces a novel approach by combining generative AI with blockchain mechanisms to address a critical challenges in vehicular security. The practical implications of this work offer substantial improvements in real-time vehicular communication systems. PrivNet demonstrates strong applicability to real-world scenarios, including secure autonomous driving, traffic coordination, and vehicle-to-infrastructure authentication. Performance analysis reveals that PrivNet outperforms state-of-the-art models, with an improvement in energy efficiency of up to 23%, improvements in detection accuracy of 12%, and a reduction in resource utilization 22%. Future work will explore the enhancement of the scalability and adaptability of this framework across diverse vehicular networks.

Data Availability Statement

This study utilised two publicly available intrusion-detection benchmarks hosted on IEEE Dataport. The first is the NSL-KDD dataset (DOI: 10.21227/425a-3e55; dataset URL: https://ieee-dataport.org/documents/nsl-kdd). The second is the CICIDS2017 dataset (DOI: 10.21227/akxq-9v09; dataset URL: https://ieee-dataport.org/documents/cicids2017).

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